Title of Lab

Your Name

Partner(s): Name

Date Submitted

Chem 2071 SectionNumber

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Introduction

In this experiment, citrus peel essential oils were isolated through steam distillation. Organics were then separated from distillate using liquid-liquid extraction. Collected oils were weighed and characterized qualitativelyy.

Materials and Methods

In a 100 mL round bottom flask, 50 mL of hot faucet water and penny-sized citrus fruit peels were added, filling the flask halfway. Thereafter, a steam distillation apparatus was constructed according to Figure 1. on page 37 of CHEM 2071 Organic Chemistry 1 Lab Manual (Keller, 2022). Heat was gradually added until boiling and the thermometer read 100°C. After collecting approximately 15 mL of distillate, the steam distillation apparatus was disassembled and cleaned with soap and water. The collected distillate was placed into a centrifuge tube along with an equal amount of ether. Afterward, both aqueous and organic layers of the centrifuge tube were vigorously shaken. The bottom aqueous layer was then extracted using a glass pipette and added to a 25 mL Erlenmeyer flask. Similarly, the remaining organic layer was poured into a separate Erlenmeyer flask of 25 mL. To the now empty centrifuge tube, the previously separated aqueous layer and another equal portion of diethyl ether was added. The extraction procedure was repeated until three organic fractions had been collected. Thereafter, sodium sulfate was used to dry organic fractions. Next, after noting the weight of a watch glass, to it was decanted all the collected organic fractions. The filled watch glass was then left to dry, revealing the essential oils. The now oil covered watch glass was weighed and qualitatively characterized. All remaining glassware was washed with soap and water.

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Results

Weight of Dry Watch Glass (g)	30.000
Weight of Oil Covered Watch Glass (g)	31.000
Weight of Essential Oils (g)	1.000

Table 1. Weight of watch glass, oil covered watch glass, and essential oils

Calculations

$$30.000 - 31.000 = 1.000\ grams\ of\ Essential\ Oils$$

Discussion

In this experiment, 1.000 g of essential oils was isolated from citrus fruit peels using steam distillation and liquid-liquid extraction. Qualitative assessment of the collected oils and its inability to evaporate away from the watch glass suggests that what was collected was the essential oils of the fruit peels. Further analysis could be taken to identify the composition of the oil: thin layer chromatography, spectroscopy, and H-NMR. Possible sources of error include insufficient drying of organic layer, inaccurate separation of organic layer from water layer during liquid-liquid extraction and weighing the oil covered watch glass before all the ether evaporated. Of the sources of error mentioned, the one to have most likely occurred is the insufficient drying of organic layer. It was unclear if the sodium sulfate drying agent gathered all the water from the organic layer. Therefore, it is possible that some of the remaining oil is water, leading to an overestimated watch glass weight. One way to resolve this error would be leave the extract to dry overnight.

References

Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2071 Organic Chemistry 1 Lab Manual [lab manual]. Auburn (Al): Auburn University.

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